MILITARY STANDARD

MIL-STD-1325-197
(NAVY)

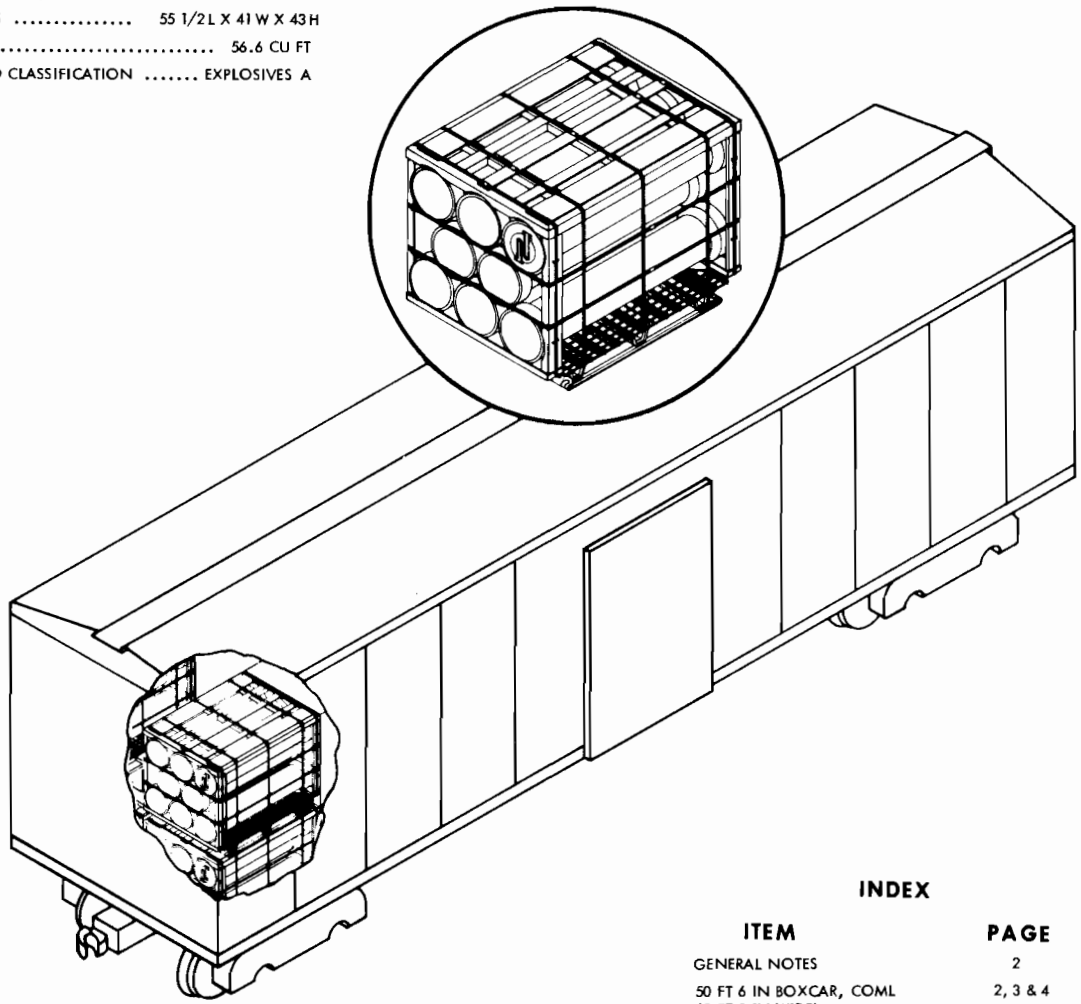
30 MAY 1980

RAILCAR LOADING OF HAZARDOUS MATERIALS

GUIDED MISSILE BGM-71A-1 (TOW)
IN ADAPTER ADU-486/E
UNIT LOAD FOR UNDERWAY REPLENISHMENT

UNIT LOAD DATA

UNIT LOAD DRAWING MIL-STD-1323-297
WEIGHT (APPROX.) ..... 1017 LBS.
DIMENSIONS ..... 55 1/2 L X 41 W X 43 H
CUBE ..... 56.6 CU FT
DOT HAZARD CLASSIFICATION ..... EXPLOSIVES A



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NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CONTAINERLOADING MILITARY STANDARDS REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY).

FSC 8140

AUTHORIZED AND RELEASED FOR GENERAL USE.

M S Gray NWHC 5/13/80
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J. E. Kelly 5/13/80
SIGNATURE AIRSYSCOM BY DIRECTION DATE

APPROVED BY BUREAU OF EXPLOSIVES

M. R. Miller 5/5/80
SIGNATURE SUPERVISOR, MILITARY & INTERMODEL SERVICES DATE

ORIGINATOR Charles DeBride 5/13/80
DATE

NAVAL WEAPONS HANDLING CENTER
WPNSTA EARLE, NEW JERSEY

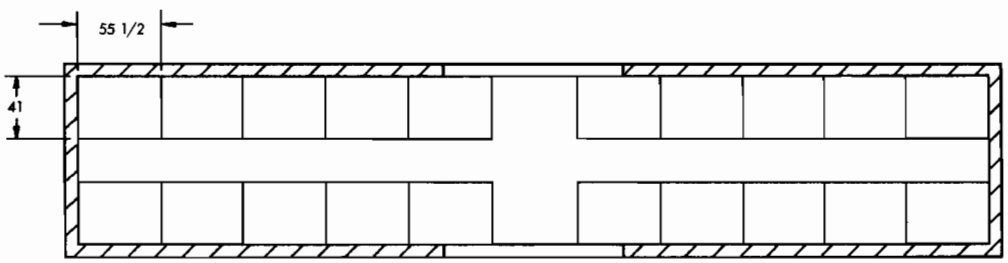
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GENERAL NOTES

1. FOR GENERAL INFORMATION CONCERNING ORDERING, INSPECTING AND PREPARING CARS, AND FOR DUNNAGING MATERIALS, DESIGN, AND INSTALLATION OF DUNNAGE SEE THE GENERAL DOCUMENT MIL-STD-1325 "RAILCAR LOADING OF HAZARDOUS MATERIALS".
2. WHEN PLANNING SHIPMENTS ORDER THE MINIMUM NUMBER OF CARS OF THE CAPACITY REQUIRED FOR THE SHIPMENT.
3. LOADING PLANS SHOWN ARE FOR COMMERCIAL BOXCARS WITH 50 FT 6 INCHES INSIDE LENGTH, 110 INCHES WIDTH, AND COMMERCIAL BOXCARS WITH 50 FT 6 INCHES LENGTH, 112 INCHES INSIDE WIDTH.
4. THE "LOAD LIMIT" OF A CAR MUST NOT BE EXCEEDED NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE "LOAD LIMIT" IS CARRIED BY ONE SET OF TRUCKS.
5. IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR.
6. THE LOAD CONSISTS OF THE CONTAINERS CNU-333/E IN ADAPTER ADU-486/E PALLETIZED IN ACCORDANCE WITH MIL-STD-1323-297.
7. THE UNIT LOADS ARE HANDLED AND LOADED WITH A SUITABLE FORK LIFT TRUCK.
8. UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325.
9. ALL STRAP JOINTS SHALL BE CRIMP-TYPE JOINTS MADE WITH A CRIMPING TOOL THAT DOES NOT CUT THE SEAL OR STRAP, RATHER THAN A TOOL THAT MAKES A NOTCH TYPE JOINT.
10. UNLESS OTHERWISE SPECIFIED, ALL MATERIAL SHALL BE AS SPECIFIED IN THE GENERAL RAILCAR LOADING DOCUMENT MIL-STD-1325.

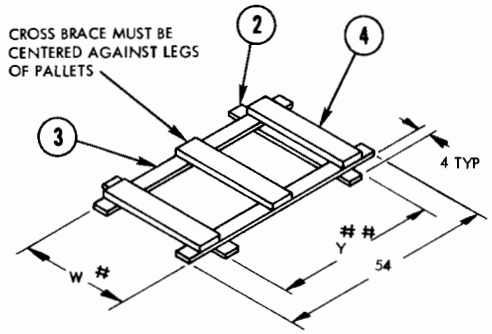
50 FT 6 IN BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF EITHER 40 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING, OR 56 UNIT LOADS WHICH MAY BE LOADED AS SHOWN IN THE ALTERNATE LOADING PLAN ON PAGE 5 WHEN A BOXCAR IS AVAILABLE WITH A 9 FT 4 INCH OR GREATER INSIDE WIDTH. ALTERNATE CENTER GATE, DETAIL E, MUST BE USED WITH THIS ALTERNATE LOADING PLAN, USING THE SAME NUMBER OF STRUTS. SWAY BRACING AND SLEEPERS ARE NOT REQUIRED BUT DOORWAY PROTECTION WILL BE REQUIRED.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH MIL-STD-1325-100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED. THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH, SEE MIL-STD-1325.
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE PROVIDING THE LENGTH OF PIECE 9 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS PROVIDING THE LENGTH OF PIECE 9 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR, AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. IF LUMBER OF SUFFICIENT LENGTH TO SPAN PLUG DOORS IS NOT AVAILABLE, RANDOM LENGTH MATERIAL, DOUBLED AND SPLICED, BUT WITH JOINTS OF SPLICES OFFSET, MAY BE USED, STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA MUST BE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4 INCH STEEL STRAPS PER STACK, EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS AND A TOP SWAY BRACE CONSISTING OF A STRAPPING BOARD AND A CROSS BRACE, TWO PER STACK, (SEE DETAIL D), MUST BE INSTALLED BETWEEN THE ROWS OF THE DOORWAY STACKS. STRAP IS SECURED TO STRAPPING BOARD WITH TWO 1 1/4 INCH STAPLES. PIECES 11 AND 12 ARE NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED, EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR, PIECES 11 AND 12 ARE REQUIRED FOR THE CONVENTIONAL DOOR. SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.



LOADING PLAN

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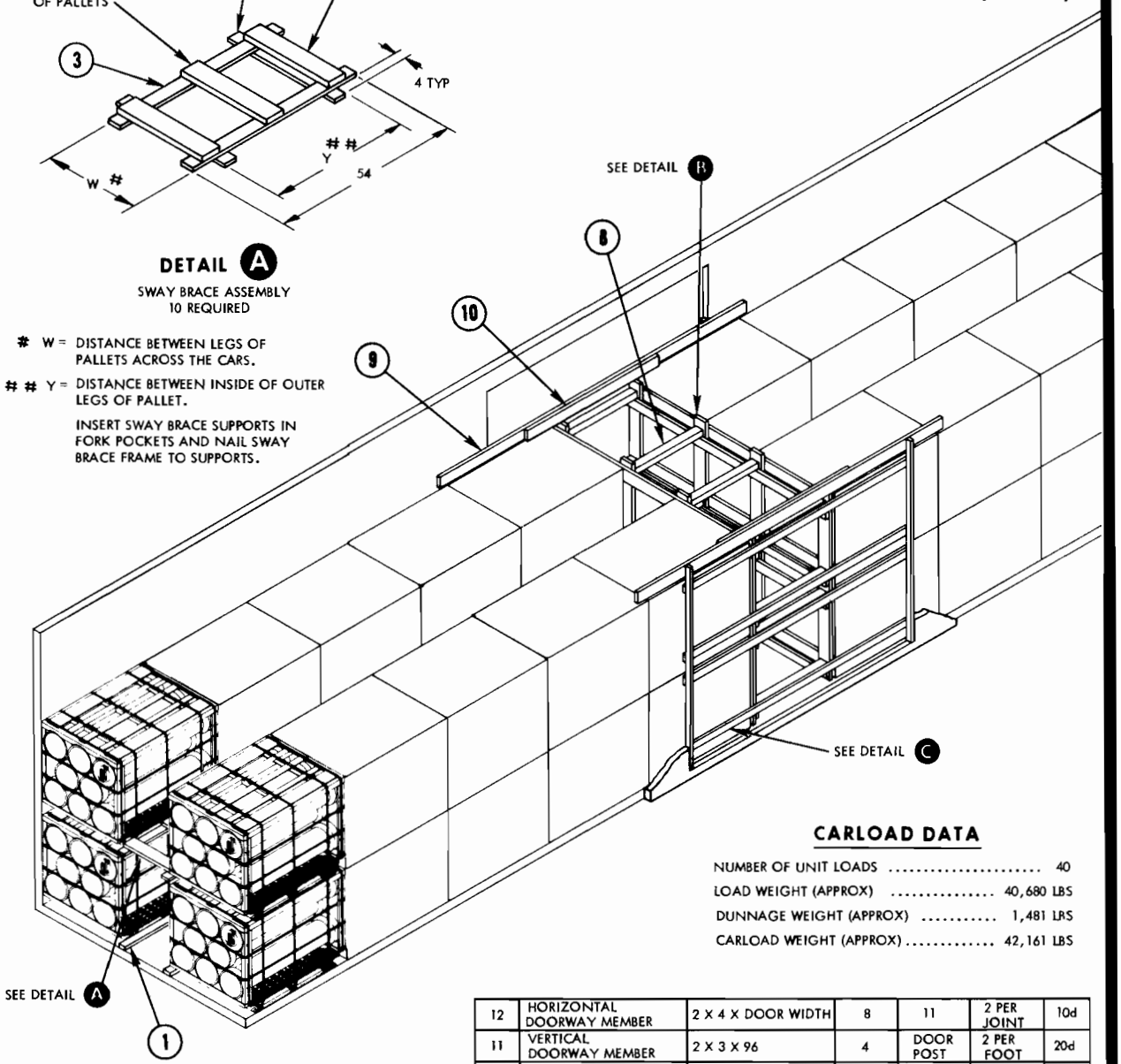
DETAIL A

SWAY BRACE ASSEMBLY
10 REQUIRED

* W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CARS.

\# \# Y = DISTANCE BETWEEN INSIDE OF OUTER LEGS OF PALLET.

INSERT SWAY BRACE SUPPORTS IN FORK POCKETS AND NAIL SWAY BRACE FRAME TO SUPPORTS.



CARLOAD DATA

NUMBER OF UNIT LOADS	40
LOAD WEIGHT (APPROX)	40,680 LBS
DUNNAGE WEIGHT (APPROX)	1,481 LBS
CARLOAD WEIGHT (APPROX)	42,161 LBS

* W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR

** PIECE 4 NAILS TO PIECE 5 WITH 10d NAILS, CLINCHED, 3 PER JOINT, AND TO PIECE 3 WITH 6d NAILS, 1 PER JOINT.

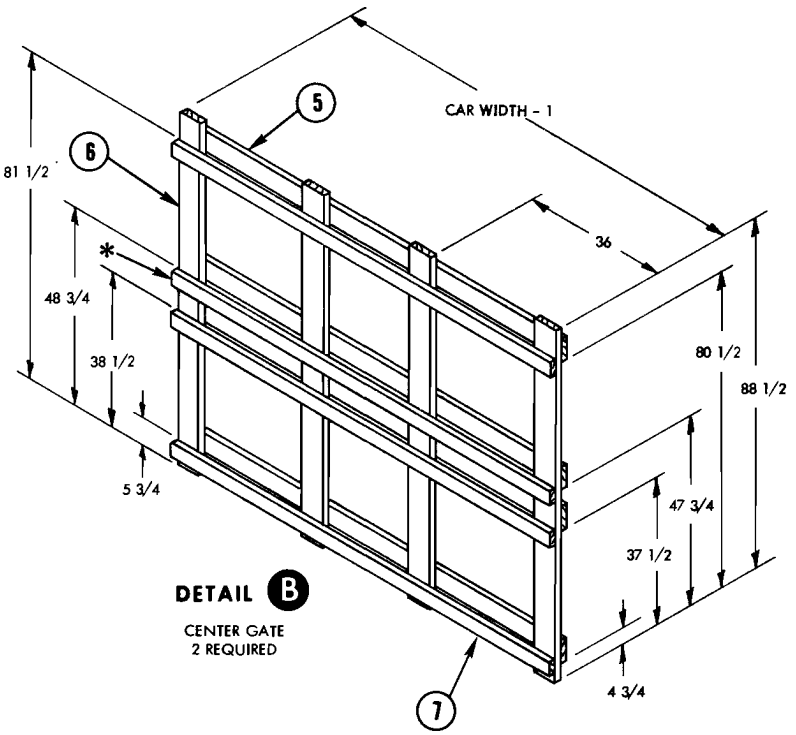
*** 2 X 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 X 4'S.

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQ'D	NAIL TO	NUMBER NAILS	SIZE NAILS
12	HORIZONTAL DOORWAY MEMBER	2 X 4 X DOOR WIDTH	8	11	2 PER JOINT	10d
11	VERTICAL DOORWAY MEMBER	2 X 3 X 96	4	DOOR POST	2 PER FOOT	20d
10	HOLD DOWN CLEAT	2 X 6 X 72	2	9	2 PER FOOT	10d
9	GATE HOLD DOWN	2 X 6 X 14 FT	2	CAR WALL	5 EACH END	10d
8	STRUT	4 X 4 X WEDGE FIT ***	16	6	2 PER JOINT	16d
7	CENTER GATE STRUT CLEAT	2 X 4 X CAR WIDTH - 1	8	6	3 PER JOINT	10d
6	CENTER GATE VERTICAL	2 X 6 X 88 1/2	8	SEE 5	-	-
5	CENTER GATE HORIZONTAL	2 X 6 X CAR WIDTH - 1	8	6	3 PER JOINT	10d
4	CROSS BRACE	2 X 6 X W *	30	SEE 3	-	-
3	SWAY BRACE LONGITUDINAL	1 X 4 X 54	20	3,5	SEE NOTE **	
2	SWAY BRACE SUPPORT	1 X 4 X (W +8) *	20	SEE 3	-	-
1	SLEEPER	2 X 4 X 54	20	CAR FLOOR	2 PER JOINT	16d

LIST OF MATERIALS & NAILING DATA

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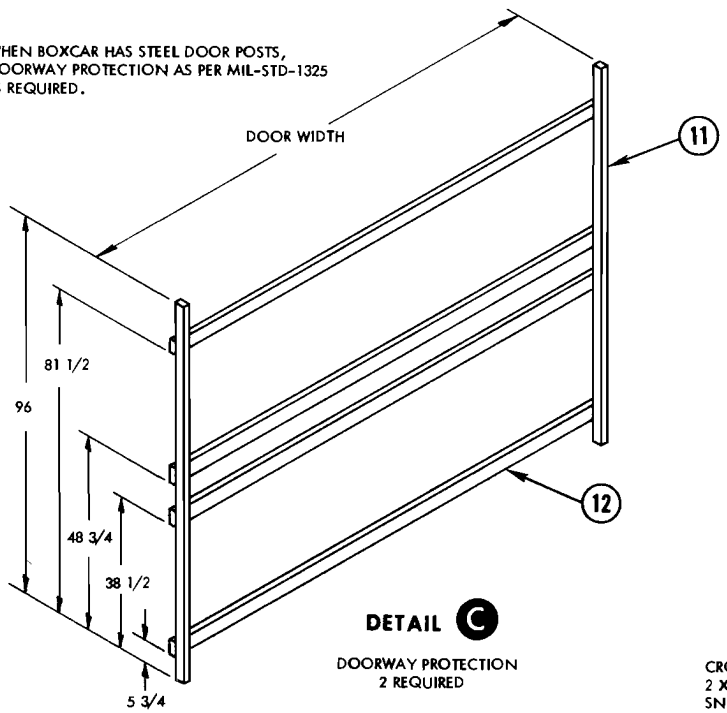
* POSITION AFTER STRUTS HAVE BEEN INSTALLED ON LOWER STRUT CLEAT (REQUIRED ONE GATE ONLY)



DETAIL B

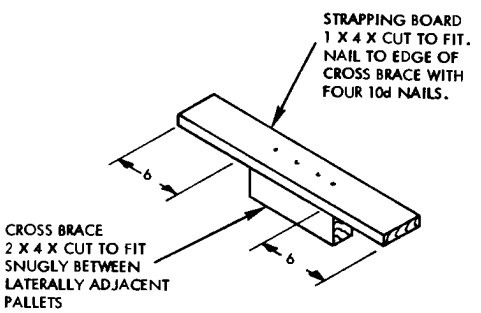
CENTER GATE
2 REQUIRED

WHEN BOXCAR HAS STEEL DOOR POSTS, DOORWAY PROTECTION AS PER MIL-STD-1325 IS REQUIRED.



DETAIL C

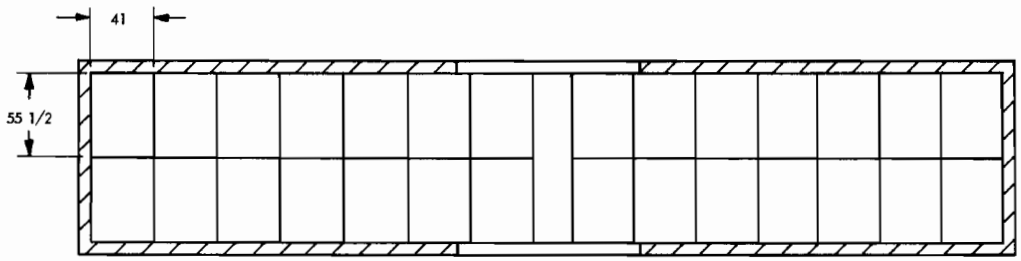
DOORWAY PROTECTION
2 REQUIRED



DETAIL D

TOP SWAY BRACE

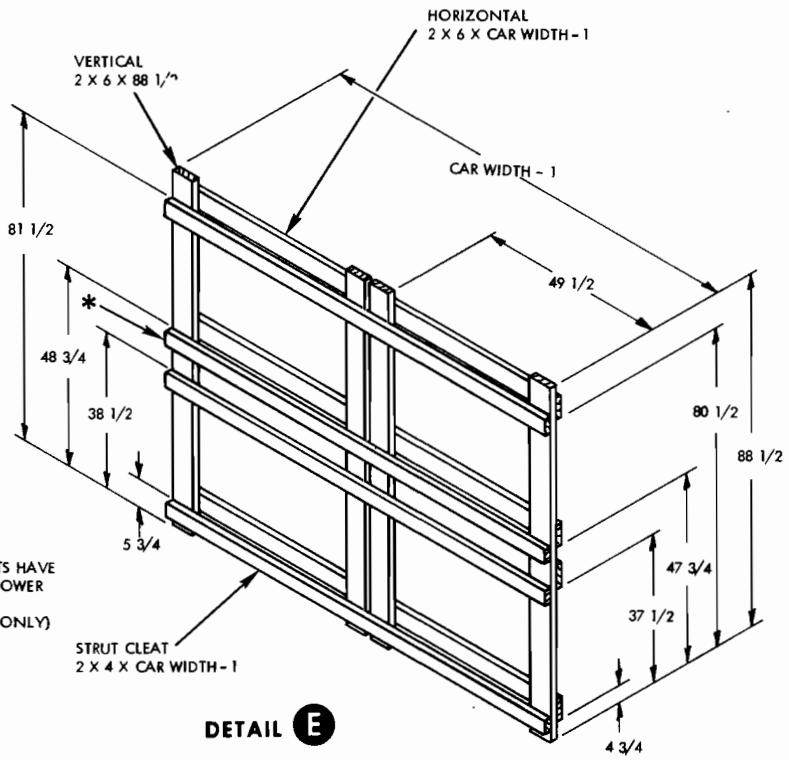
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ALTERNATE LOADING PLAN
FOR RAILCAR 9 FT. 4 INCHES INSIDE WIDTH OR GREATER

ALTERNATE CARLOAD DATA

NUMBER OF UNIT LOADS	56
LOAD WEIGHT	56,952 LBS.
DUNNAGE WEIGHT	785 LBS.
CARLOAD WEIGHT	57,737 LBS.



* POSITION AFTER STRUTS HAVE BEEN INSTALLED ON LOWER STRUT CLEAT (REQUIRED ONE GATE ONLY)

DETAIL E

ALTERNATE CENTER GATE
2 REQUIRED

REVIEW ACTIVITY :
NAVY - OS, AS

PREPARING ACTIVITY :
NAVY - OS
(PROJECT NO. 8140 - N477)

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